

ORIGINAL ARTICLE

Elevated Endogenous Insulin and Insulin Resistance Are Associated With Progressive Diastolic Dysfunction and Worse Cardiovascular Outcomes

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BACKGROUND: Although diabetes is a known driver of heart failure with preserved ejection fraction, whether antecedent hyperinsulinemia drives preclinical diastolic dysfunction (DD) progression before overt hyperglycemia remains unknown. Therefore, we investigated whether elevated endogenous insulin and insulin resistance independently predict longitudinal DD progression and adverse cardiovascular outcomes.

METHODS: In a community-based cohort study, we evaluated 1191 participants from the Olmsted County Heart Function Study (median age, 58.7 [interquartile range, 52.4–66.6 years]) who underwent echocardiography at baseline and after 4 years. Fasting insulin and homeostatic model assessment of insulin resistance were assessed. The primary end point was DD progression. Secondary outcomes (mortality, heart failure) were assessed in 2030 participants followed for a median of 10.97 (95% CI, 10.90–11.05) years for heart failure and a median of 18.1 (95% CI, 18.0–18.2) years for mortality.

RESULTS: DD progression occurred in 7.5% of participants. Baseline fasting insulin and homeostatic model assessment of insulin resistance strongly predicted progression ($P < 0.001$). In multivariable analysis, every 1-SD increase in log-transformed insulin was associated with 65% increased odds of progression (odds ratio, 1.65 [95% CI, 1.33–2.07]; $P < 0.001$), adjusting for age, creatinine, and NT-proBNP (N-terminal pro-B-type natriuretic peptide). Similar results were observed for homeostatic model assessment of insulin resistance (odds ratio, 1.66 [95% CI, 1.34–2.08]; $P < 0.001$). In secondary analyses, elevated insulin independently predicted all-cause (hazard ratio, 1.11 per 1-SD [95% CI, 1.02–1.21]) and cardiovascular mortality (hazard ratio, 1.16 [95% CI, 1.02–1.33]). Homeostatic model assessment of insulin resistance was similarly predictive of all-cause (hazard ratio, 1.11 [95% CI, 1.03–1.20]) and cardiovascular mortality (hazard ratio, 1.16 [95% CI, 1.02–1.32]).

CONCLUSIONS: Elevated endogenous insulin and insulin resistance are predictors of progressive DD and long-term mortality, even after accounting for body mass index. Metabolic dysregulation may drive myocardial stiffening before diabetes onset, identifying a potential early therapeutic window for heart failure with preserved ejection fraction prevention.

Key Words: body mass index ■ creatinine ■ heart failure ■ insulin ■ prevalence

Patients with diabetes are known to have a higher prevalence of diastolic dysfunction (DD) and are at increased risk for developing heart failure with preserved ejection fraction (HFpEF).^{1–3} Although the relationship between overt diabetes and diastolic abnormalities

is well described,³ the impact of more subtle metabolic disturbances, such as elevated endogenous insulin levels or insulin resistance, on the progression of DD remains less clear. These metabolic parameters, often quantified by fasting insulin and the homeostatic model assessment

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WHAT IS NEW?

- In a community-based cohort, elevated baseline fasting insulin and homeostatic model assessment of insulin resistance independently predicted progression of diastolic dysfunction on paired echocardiography, while fasting glucose was still normal.
- The same baseline metabolic measurement that predicted subclinical diastolic progression also predicted long-term all-cause and cardiovascular mortality in the same cohort.
- Among participants with obesity, low rather than high fasting insulin conferred the greatest heart failure risk, likely consistent with beta-cell exhaustion marking more advanced metabolic disease.

WHAT ARE THE CLINICAL IMPLICATIONS?

- Fasting insulin and homeostatic model assessment of insulin resistance are inexpensive and widely available and may identify patients at risk for diastolic progression and heart failure before glucose or HbA1c becomes abnormal.
- The subclinical, normoglycemic, hyperinsulinemic phase represents a candidate therapeutic window during which insulin-sensitizing strategies might alter the trajectory toward heart failure with preserved ejection fraction.
- In patients with obesity or insulin resistance, a normal or low natriuretic peptide should be read with caution, and a low fasting insulin may signal failing secretion rather than metabolic health.

Nonstandard Abbreviations and Acronyms

BMI	body mass index
DD	diastolic dysfunction
HFpEF	heart failure with preserved ejection fraction
HOMA-IR	homeostatic model assessment of insulin resistance
HR	hazard ratio
NT-proBNP	N-terminal pro-B-type natriuretic peptide
OCHF	Olmsted County Heart Function Study
OR	odds ratio
SGLT2i	sodium-glucose cotransporter-2 inhibitor

of insulin resistance (HOMA-IR),⁴ reflect early pathophysiologic changes that precede overt hyperglycemia.^{5,6} Several studies support a mechanistic link between insulin resistance and myocardial remodeling, demonstrating that excess insulin signaling can drive interstitial fibrosis, increase myocardial stiffness, and promote adverse remodeling.^{7,8} Yet, to our knowledge, whether

these metabolic factors predict worsening diastolic function over time in community populations remains to be determined.

Further, there is growing interest in understanding how insulin resistance and hyperinsulinemia relate to long-term cardiovascular outcomes. Population-based studies have linked HOMA-IR and related indices to incident heart failure⁹ and mortality,¹⁰ but most have not included direct assessment of DD or progression.

Understanding the relationships between these metabolic markers, progression of DD, and adverse clinical outcomes may have important clinical implications. Identifying patients with subclinical insulin resistance who are at risk for progressive DD may allow for earlier preventive strategies. Furthermore, therapies that improve plasma glycemic milieu, such as SGLT2i (sodium-glucose cotransporter-2 inhibitors)¹¹ and structured lifestyle interventions,^{12,13} have shown favorable effects on cardiovascular outcomes^{11,14} and may hold potential to modify disease trajectory.

Accordingly, we evaluated in a community-based cohort whether elevated endogenous insulin levels and insulin resistance (by HOMA-IR) are independently associated with progression of DD and with adverse clinical outcomes, including all-cause mortality, cardiovascular mortality, and heart failure. We hypothesized that higher fasting endogenous insulin levels and greater HOMA-IR indices would be associated with increased risk of progressive DD and worse cardiovascular outcomes after adjustment for conventional risk factors.

METHODS

Population, Setting, and Variable Measurement

The Mayo Clinic institutional review board granted this study exempt status. Participants provided written informed consent and research authorization at enrollment. The data that support the findings of this study are available from the corresponding author on reasonable request. The OCHF (Olmsted County Heart Function Study) cohort study was conducted in Olmsted County, Minnesota, among residents above the age of 45 and has been extensively described.^{15,16} Briefly, the OCHF enrolled a randomly selected, age- and sex-stratified sample of 2042 community-dwelling residents of Olmsted County, MN, aged 45 years or older between June 1997 and September 2000. During this initial visit, participants underwent a standardized clinical evaluation including medical history, physical examination, blood pressure measurement, and a comprehensive 2-dimensional and Doppler echocardiogram, along with the collection of fasting blood for biomarker assessment. Patients returned for a second study visit ≈4 years later, during which another echocardiogram was conducted. Major clinical outcomes (eg, all-cause mortality, cardiovascular mortality, and heart failure) were extracted electronically during follow-up.¹⁷ The primary analysis, which evaluated the association between progression of DD and baseline insulin and HOMA-IR levels, required patients to complete echocardiograms at both study visits. Of the 2042 enrolled participants, 640 were excluded for

attending only a single visit. Among the 1402 who returned for the second visit, 185 were excluded for indeterminate or incomplete diastolic function data at either visit, and 3 were excluded for grade III DD at baseline. Of the remaining 1214 participants with normal through grade II diastolic function at baseline, 23 were excluded for insulin-dependent diabetes at either visit ($n=7$), missing baseline fasting insulin ($n=2$), or having atrial fibrillation, flutter, or a paced rhythm ($n=14$), yielding a final primary analytic cohort of 1191 participants (Figure 1A). The secondary analysis, which evaluated the association between clinical outcomes and insulin and HOMA-IR, required only that patients attend the first visit, did not present with insulin-dependent diabetes, and had available insulin data. For the secondary analysis, of the 2042 enrolled participants, 12 were excluded for insulin-dependent diabetes at the baseline visit ($n=6$) or missing baseline fasting insulin ($n=6$), yielding a final

secondary analytic cohort of 2030 participants (Figure 1B). Participants were not excluded on the basis of systolic function; however, consistent with prior reports from this cohort, reduced left ventricular ejection fraction ($<50\%$) was uncommon, such that the cohort was composed predominantly of individuals with preserved ejection fraction.¹⁸

Presence and severity of DD were summarized at both study visits according to the 2016 American Society of Echocardiography/European Association of Cardiovascular Imaging definition.¹⁹ A subset of participants exhibited discordant parameters in which the criteria for advancing from grade I to II were partially but not completely met; rather than classifying these participants as indeterminate, we assigned them to an intermediate category designated grade I to II, positioned between grade I and grade II on the ordinal scale. Diastolic function was therefore ranked along the following

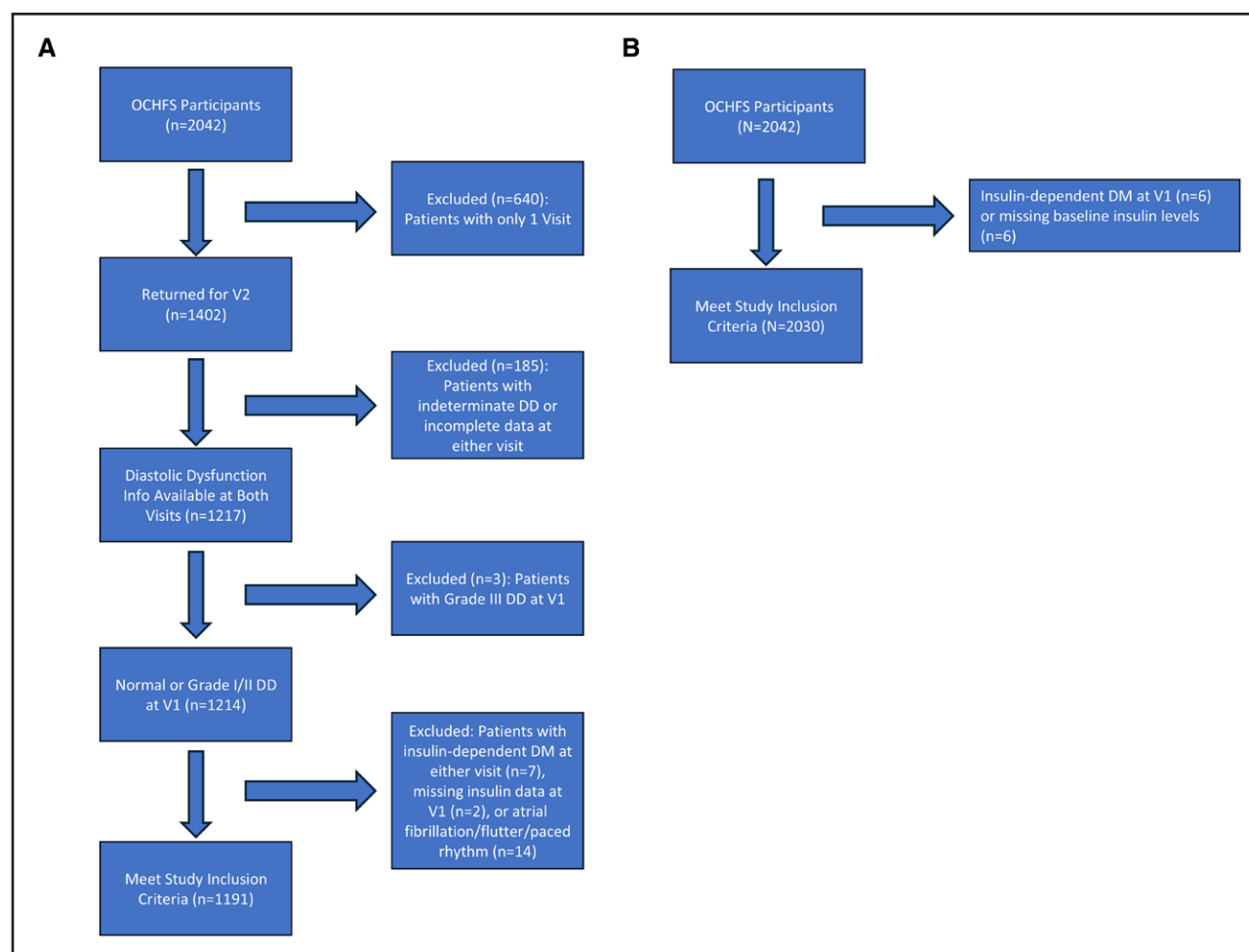


Figure 1. Patient selection flowcharts for primary and secondary analyses.

A, Primary analysis cohort ($n=1191$) for the longitudinal assessment of diastolic dysfunction (DD) progression. Among 2042 OCHFS (Olmsted County Heart Function Study) participants, 640 were excluded for having only a single study visit. Of the 1402 participants who returned for visit 2 (V2), 185 were excluded due to indeterminate diastolic function or incomplete diastolic data at either visit. Of the remaining 1217 participants with diastolic function data available at both visits, 3 were excluded for grade III DD at baseline (visit 1 [V1]). Among the remaining 1214 participants, an additional 23 were excluded for having insulin-dependent diabetes (DM) at either visit ($n=7$), missing fasting insulin data at V1 ($n=2$), or atrial fibrillation, flutter, or a paced rhythm precluding reliable diastolic grading ($n=14$), yielding a final analytic cohort of 1191 participants. Indeterminate diastolic function reflects the inability to assign a definitive diastolic grade according to American Society of Echocardiography/European Association of Cardiovascular Imaging guidelines due to discordant or insufficient echocardiographic parameters.

B, Secondary analysis cohort ($n=2030$) for the evaluation of clinical outcomes. Of the 2042 OCHFS participants, 12 were excluded for insulin-dependent DM at V1 ($n=6$) or missing baseline fasting insulin ($n=6$), resulting in a final cohort of 2030 participants.

sequence: normal, grade I, grade I to II, grade II, and grade III. Patients whose DD grade increased by at least 1 step along this sequence between visit 1 and visit 2 (including those who moved from normal diastolic function into any grade of DD, or from grade I into grade I to II) were categorized as DD progressors. Although echocardiographic parameters of diastolic function, such as tissue Doppler imaging, were not yet in widespread clinical use during this era, they were an integral component of the OCHF protocol.²⁰ HOMA-IR was computed as defined previously.⁴ Hypertension was ascertained by trained-nurse abstraction of the community medical record and defined according to the criteria of the Sixth Report of the Joint National Committee, consistent with prior reports from this cohort.¹⁸

Statistical Analysis

In descriptive analyses, categorical variables were tabulated as count (percent) while continuous variables were summarized as median (interquartile range). Comparisons of demographics, comorbidities, and biomarkers between insulin tertile groups used the Kruskal-Wallis test or Pearson χ^2 test. Rates of progression of DD were compared between insulin tertiles using the Cochran-Armitage test, while natural log-transformed continuous insulin values were compared between DD progressors and nonprogressors using the Wilcoxon rank-sum test. Logistic regression models were fit to evaluate the association between insulin and DD progression, adjusting individually for key cardiovascular risk factors. For these models, insulin was natural log-transformed and standardized to mean zero and unit variance to allow reporting of odds ratios (ORs) per 1 SD. Similar analyses were repeated using HOMA-IR instead of insulin as the independent variable.

The association between insulin and HOMA-IR and clinical outcomes was evaluated using Cox regression models adjusted for age, sex, and body mass index (BMI). Mortality data were extracted through the end of 2017, and patients were censored at the last clinical follow-up between their baseline study visit and this cutoff date. Heart failure data were available through the end of 2010, and patients were censored at the last clinical follow-up between their baseline study visit and this cutoff date. Reverse-Kaplan-Meier estimates were used to summarize median follow-up for mortality and heart failure outcomes. Results were summarized using hazard ratios (HR) and corresponding 95% CIs and *P* values. Similar analyses were conducted for HOMA-IR. Missing data were not imputed; analyses were performed on complete cases. All statistical analyses were performed in R (version 4.2.2).

RESULTS

Baseline Characteristics

The primary analysis consisted of *n*=1191 patients, as detailed in Figure 1A. In this cohort, the median (Q1–Q3) age was 58.7 years (52.4–66.6) with 49.8% females (*n*=593) and 98.2% White (*n*=1169). The median BMI (kg/m²) was in the overweight range, 27.4 (24.9–30.9). Hypertension was present in 22.8% of patients (*n*=271) and noninsulin-dependent diabetes only in 5% (*n*=59). The prevalence of coronary artery disease was 8.6% (*n*=102). About 48.7% (*n*=579) were current or former

smokers, and only 16.5% (*n*=181) were on antilipemic therapy. Median fasting serum glucose (mg/dL) was normal at 93 (88–99), and median fasting insulin (μ U/mL) was 5.0 (3.5–7.5).

Median serum creatinine (mg/dL) of 0.8 (0.7–1.0) was within normal limits, while median NT-proBNP (N-terminal pro-B-type natriuretic peptide; pg/mL) was low at 54.4 (23.0–112.5). Median total cholesterol (mg/dL) was elevated at 202 (181–225), median high-density lipoprotein (mg/dL) was 44 (36–55), and median triglycerides (mg/dL) were 125 (91–172). The median ejection fraction (%) was intact at 65 (60–68).

When stratified by baseline insulin tertiles, there were significant differences in clinical and laboratory parameters among participants (Table 1). Patients in the higher insulin tertiles were generally older, had higher body mass indices, and were less often female, with a greater prevalence of hypertension. Similarly, the prevalence of coronary artery disease, diabetes, and use of antilipemic therapy increased with increasing tertiles. In terms of laboratory values, higher tertiles were associated with higher fasting serum glucose, lower NT-proBNP levels, lower high-density lipoprotein cholesterol, and higher triglycerides. There were no significant differences in race, smoking status, or left ventricular ejection fraction.

Baseline Insulin and DD Progression

We evaluated the progression of DD within each baseline insulin tertile. DD progression was defined as an increase of at least 1 grade between visit 1 and visit 2 along the following sequence: normal, grade I DD, grade I to II, grade II DD, and grade III DD (Figure 2). Overall, 89 patients (7.5%) showed DD progression. The proportion of participants with DD progression increased stepwise across insulin tertiles (Figure 3A; *P*<0.001), from 3.5% in tertile 1, to 7.3% in tertile 2, and 11.6% in tertile 3. Likewise, when comparing baseline insulin levels as a continuous (log-transformed) variable between participants with and without DD progression, those with progression had higher baseline insulin levels (Figure 3B; *P*<0.001).

Seven adjusted logistic regression models were fit, each adjusting for one of the following: age, BMI, hypertension, coronary artery disease, antilipemic therapy, creatinine, and NT-proBNP. Insulin remained significantly associated with DD progression in all models. Adjusted for age, insulin had a per SD OR of 1.48 (95% CI, 1.20–1.83). Adjusted for BMI, the estimated OR was 1.38 (95% CI, 1.08–1.76; Table 2). When age and BMI were included simultaneously, the effect of insulin was diminished (OR, 1.22 [95% CI, 0.95–1.55]), although the point estimate continued to suggest a positive trend. Notably, in a model incorporating age along with creatinine and NT-proBNP, insulin remained independently associated with DD progression (OR, 1.65 [95% CI, 1.33–2.07]). Further, in a sensitivity analysis excluding all participants

Table 1. Baseline Characteristics by Insulin Tertiles

Characteristic	Insulin tertile 1 (n=398)	Insulin tertile 2 (n=396)	Insulin tertile 3 (n=397)	Total (n=1191)	P value
Demographics					
Age at exam, y	57.2 (51.2–64.5)	59.3 (52.0–66.5)	60.2 (54.0–67.5)	58.7 (52.4–66.6)	<0.001
Female	235 (59.0)	184 (46.5)	174 (43.8)	593 (49.8)	<0.001
White	390 (98.0)	390 (98.5)	389 (98.0)	1169 (98.2)	0.835
Clinical characteristics					
BMI, kg/m ²	25.2 (22.9–27.0)	27.5 (25.2–30.1)	30.8 (27.7–34.3)	27.4 (24.9–30.9)	<0.001
Hypertension	54 (13.6)	87 (22.0)	130 (32.7)	271 (22.8)	<0.001
Coronary artery disease	24 (6.0)	28 (7.1)	50 (12.6)	102 (8.6)	0.002
Diabetes	9 (2.3)	12 (3.0)	38 (9.6)	59 (5.0)	<0.001
Current/former smoker*	184 (46.2)	199 (50.4)	196 (49.6)	579 (48.7)	0.461
Medications					
Antilipemic therapy*	42 (11.4)	49 (13.6)	90 (24.5)	181 (16.5)	<0.001
Laboratory values					
Serum glucose, mg/dL	90.0 (85.0–94.0)	94.0 (89.0–99.0)	97.0 (91.0–106.0)	93.0 (88.0–99.0)	<0.001
Insulin, μ U/mL	3.0 (2.4–3.5)	5.0 (4.4–5.7)	9.2 (7.5–12.3)	5.0 (3.5–7.5)	<0.001
Range	0.8–3.9	3.91–6.55	6.56–82.7	0.8–82.7	
HOMA-IR	0.6 (0.5–0.8)	1.1 (1.0–1.3)	2.0 (1.7–2.6)	1.1 (0.8–1.6)	<0.001
Range	0.2–1.0	0.8–1.7	1.3–8.7	0.2–8.7	
Creatinine, mg/dL*	0.8 (0.7–0.9)	0.8 (0.7–0.9)	0.9 (0.7–1.0)	0.8 (0.7–1.0)	<0.001
NT-proBNP, pg/mL*	70.5 (32.3–129.4)	54.4 (22.1–108.9)	40.3 (16.4–100.0)	54.4 (23.0–112.5)	<0.001
Total cholesterol, mg/dL*	200.0 (181.0–221.0)	207.0 (185.0–231.2)	200.0 (176.0–222.0)	202.0 (181.0–225.0)	0.005
HDL cholesterol, mg/dL*	53.0 (42.0–63.0)	44.0 (37.0–54.0)	37.0 (31.0–46.0)	44.0 (36.0–55.0)	<0.001
Triglycerides, mg/dL	102.5 (77.0–130.0)	128.0 (91.8–174.0)	153.0 (113.0–210.0)	125.0 (91.0–172.0)	<0.001
Echocardiography					
LVEF, %	65.0 (60.0–67.0)	63.0 (60.0–65.0)	65.0 (60.0–68.0)	65.0 (60.0–68.0)	0.113
E/A*	1.2 (1.0–1.5)	1.1 (0.9–1.3)	1.0 (0.8–1.2)	1.1 (0.9–1.4)	<0.001
E/e*	7.3 (6.2–8.8)	7.8 (6.2–10.0)	8.6 (6.7–10.0)	7.8 (6.4–10.0)	<0.001
E'	9.0 (7.0–11.0)	8.0 (7.0–10.0)	8.0 (6.0–9.0)	8.0 (7.0–10.0)	<0.001
TR velocity*	2.3 (2.2–2.4)	2.3 (2.2–2.5)	2.4 (2.2–2.5)	2.3 (2.2–2.5)	0.056

Values are n (%) or median (Q1–Q3). BMI indicates body mass index; E/A, ratio of early (E) to late (A) mitral inflow velocity; E/e'; ratio of early mitral inflow velocity to early diastolic mitral annular velocity; E'; early diastolic mitral annular velocity; HDL, high-density lipoprotein; HOMA-IR, homeostatic model assessment of insulin resistance; LVEF, left ventricular ejection fraction; NT-proBNP, N-terminal pro-B-type natriuretic peptide; Q1, first quartile; Q3, third quartile; and TR, tricuspid regurgitation.

*Missing data, n (tertile 1/2/3): smoking status, 3 (0/1/2); antilipemic therapy, 95 (30/35/30); creatinine, 37 (7/19/11); NT-proBNP, 26 (7/10/9); total cholesterol, 1 (1/0/0); HDL cholesterol, 1 (1/0/0); E/A, 6 (0/3/3); E/e', 165 (33/64/68); E', 163 (33/62/68); TR velocity, 360 (70/123/167). P values from the Kruskal-Wallis test (continuous variables) or χ^2 test (categorical variables).

with non-insulin-dependent diabetes at either visit (n=111; remaining n=1080), DD progression again increased stepwise across insulin tertiles (3.3%, 5.8%, 10.5%; $P<0.001$; Figure S1).

Baseline HOMA-IR and DD Progression

We next examined baseline insulin resistance using HOMA-IR. Participants were categorized into 3 groups based on baseline HOMA-IR (<1 , 1 – 1.99 , and ≥ 2). The prevalence of DD progression also increased stepwise across categories, from 3.5% to 9.1% and 12.0%, respectively ($P<0.001$; Figure 4A). Similarly, when HOMA-IR was analyzed as a continuous log-transformed variable, participants with DD progression had significantly higher

baseline HOMA-IR values compared with those without progression (Figure 4B; $P<0.001$).

Higher baseline HOMA-IR was consistently associated with increased odds of DD progression across multiple models. The association remained statistically significant after adjusting individually for key clinical covariates, including age (OR, 1.48 [95% CI, 1.20–1.82]) and BMI (OR, 1.41 [95% CI, 1.11–1.77]), and was robust to adjustment for hypertension, coronary artery disease, antilipemic therapy, creatinine, and NT-proBNP (Table 2). In a model that included both age and BMI simultaneously, the effect size was attenuated (OR, 1.22 [95% CI, 0.96–1.55]), although the trend remained directionally positive. Importantly, in a model adjusting for age, creatinine, and NT-proBNP, HOMA-IR retained an

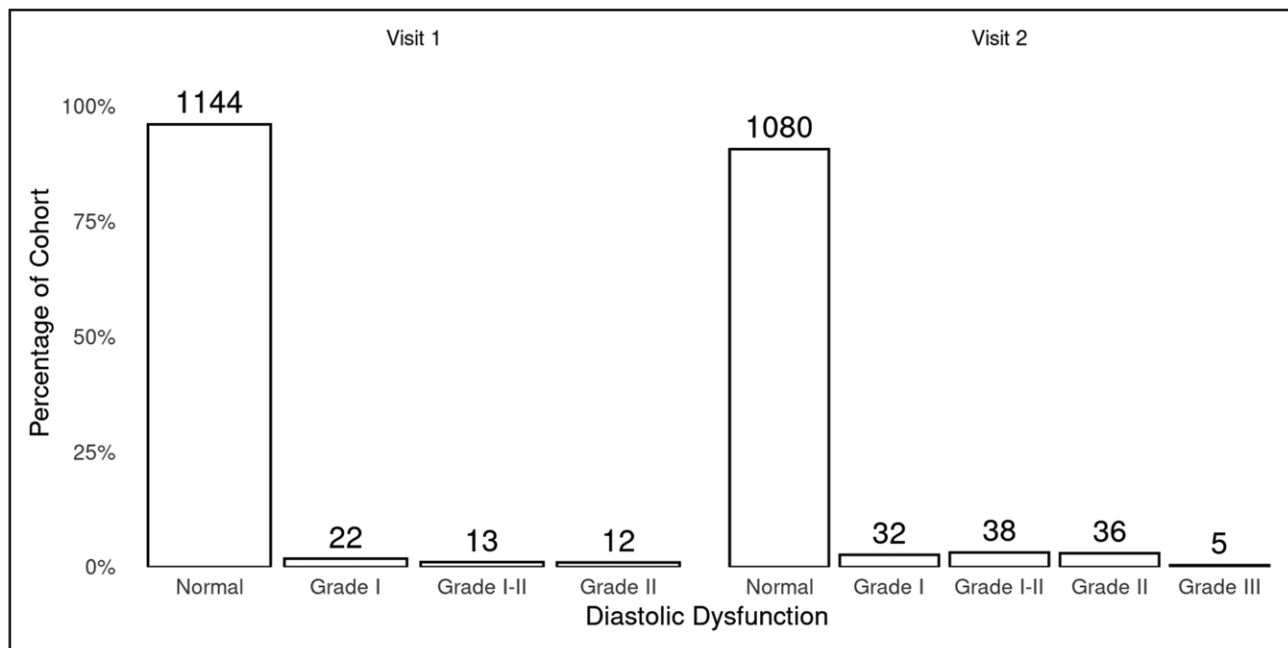


Figure 2. Diastolic dysfunction (DD) severity at visit 1 and visit 2.

Bar charts depicting the distribution of DD grade in the primary analytic cohort ($n=1191$) at the baseline visit (visit 1) and the follow-up visit (visit 2) occurring ≈ 4 years later, classified according to the 2016 American Society of Echocardiography/European Association of Cardiovascular Imaging definition. The y-axis represents the percentage of the cohort within each grade, and counts are displayed above each bar. At visit 1, the cohort was composed of normal diastolic function ($n=1144$), grade I ($n=22$), grade I-II ($n=13$), and grade II ($n=12$); participants with grade III DD at baseline were excluded by design. At visit 2, the distribution shifted toward higher grades, with normal ($n=1080$), grade I ($n=32$), grade I-II ($n=38$), grade II ($n=36$), and grade III ($n=5$), reflecting net progression of diastolic dysfunction over the follow-up interval.

independent association with DD progression (OR, 1.66 [95% CI, 1.34–2.08]).

Baseline Insulin and HOMA-IR and Clinical Outcomes

The secondary analysis consisted of $N=2030$ patients (Figure 1B; Table S1). In this cohort, 589 deaths were observed, of which 188 were attributed to cardiac-related causes. Further, 254 patients experienced heart failure. The median follow-up was 10.97 (95% CI, 10.90–11.05) years for heart failure and 18.1 (95% CI, 18.0–18.2) years for mortality.

In models adjusted for age, sex, and BMI, both fasting insulin and HOMA-IR were significantly associated with all-cause and cardiac mortality. Specifically, each 1-SD increase in insulin was associated with a higher risk of all-cause mortality (HR, 1.11 [95% CI, 1.02–1.21]; $P=0.012$) and cardiovascular mortality (HR, 1.16 [95% CI, 1.02–1.33]; $P=0.026$). Similar associations were observed for HOMA-IR, with each 1-SD increase corresponding to higher risks of all-cause mortality ($P=0.009$) and cardiovascular mortality ($P=0.022$; Table 3).

Significant interaction was observed between insulin resistance and BMI for the heart failure outcome. Even when stratified by BMI, insulin resistance exhibited a nonlinear relationship with heart failure. Accordingly,

insulin and HOMA-IR were categorized as normal if they were within one SD of the mean on the log scale and low or high if they fell outside this range (Table 4). Within the nonobese strata, higher insulin and HOMA-IR were associated with increased heart failure risk, consistent with insulin resistance: high insulin predicted heart failure in participants with BMI <26 (HR, 2.28 [95% CI, 1.09–4.81]), and high HOMA-IR did so in both the BMI <26 (HR, 2.37 [95% CI, 1.13–4.99]) and BMI 26 to 29.9 (HR, 2.15 [95% CI, 1.14–4.06]) strata. On the contrary, among obese participants (BMI ≥ 30), the pattern reversed, with low insulin rather than high conferring the greatest risk in the entire analysis (HR, 3.93 [95% CI, 1.41–10.92]).

DISCUSSION

In this community-based longitudinal study, we report 3 novel and clinically relevant findings regarding the impact of metabolic dysregulation on myocardial structure and function. First, we demonstrated that elevated endogenous insulin levels and insulin resistance, as measured by HOMA-IR, are independently associated with the longitudinal progression of DD over 4 years. This relationship persisted even after rigorous adjustment for traditional cardiovascular risk factors, including blood pressure, BMI, and baseline natriuretic peptide

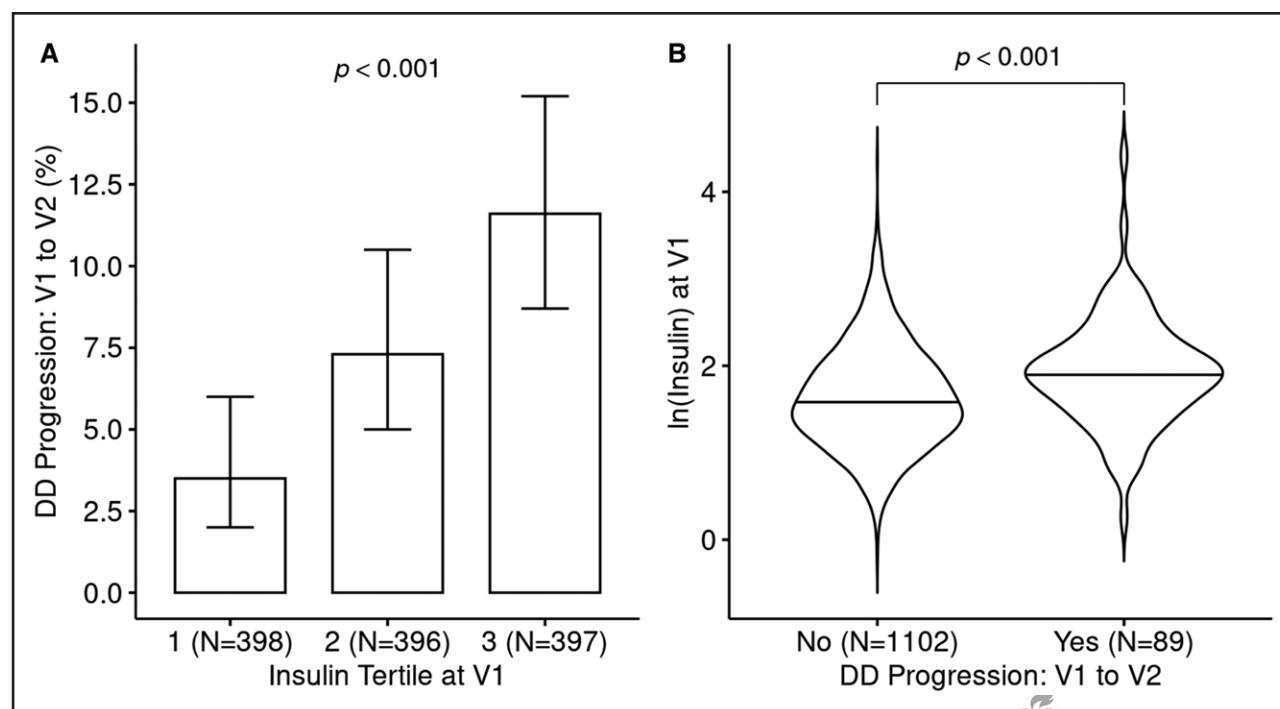


Figure 3. Association between baseline insulin and progression of diastolic dysfunction (DD).

A, Bar graph showing the proportion of participants with progression of DD by insulin tertile, with 95% CIs displayed using error bars. DD progression was defined as an increase of at least 1 grade between the first and second echocardiographic assessments (normal, grade I, grade I–II, grade II, grade III). Progression occurred in 14/398 (3.5%) of tertile 1, 29/396 (7.3%) of tertile 2, and 46/397 (11.6%) of tertile 3 ($P < 0.001$; Cochran-Armitage test). **B**, Violin plot showing distribution of baseline natural log-transformed insulin levels according to progression status, with higher levels observed in participants whose DD progressed ($P < 0.001$; Wilcoxon test).

levels. Second, we found that this risk is graded; higher tertiles of insulin and HOMA-IR were associated with a stepwise increase in the likelihood of worsening diastolic function. Third, in a broader secondary analysis of over 2000 community-dwelling adults, these metabolic markers were significantly associated with an increased risk of long-term all-cause and cardiovascular mortality. Collectively, these findings suggest that elevated baseline insulin levels and insulin resistance are not merely markers of metabolic syndrome but may be direct

drivers of myocardial stiffening and early heart failure pathogenesis, occurring well before the onset of overt hyperglycemia.

To the best of our knowledge, no prior study unites graded diastolic progression with hard outcomes in 1 community cohort. Earlier longitudinal work tracked drift in continuous indices rather than transitions across graded diastolic categories (ULSAM,²¹ using peak E and A velocities and left atrial diameter from a single late-life echocardiogram; FLEMENGHO,²² a smaller family based sample

Table 2. Association of Baseline Insulin and HOMA-IR Index with Progression of Diastolic Dysfunction in Adjusted Models

Model	OR per 1-SD increase in log (insulin; 95% CI)	P value	OR per 1-SD increase in log (HOMA-IR; 95% CI)	P value
Adjusted for age	1.48 (1.20–1.83)	<0.001	1.48 (1.20–1.82)	<0.001
Adjusted for BMI	1.38 (1.08–1.76)	0.009	1.41 (1.11–1.77)	0.004
Adjusted for hypertension	1.42 (1.15–1.76)	<0.001	1.43 (1.17–1.75)	<0.001
Adjusted for CAD	1.43 (1.16–1.76)	<0.001	1.43 (1.17–1.75)	<0.001
Adjusted for antilipemic therapy	1.45 (1.17–1.79)	<0.001	1.45 (1.18–1.78)	<0.001
Adjusted for creatinine	1.53 (1.24–1.88)	<0.001	1.54 (1.26–1.88)	<0.001
Adjusted for NT-proBNP	1.75 (1.42–2.18)	<0.001	1.77 (1.44–2.19)	<0.001
Adjusted for age + BMI	1.22 (0.95–1.55)	0.116	1.22 (0.96–1.55)	0.094
Adjusted for age + creatinine + NT-proBNP	1.65 (1.33–2.07)	<0.001	1.66 (1.34–2.08)	<0.001

BMI indicates body mass index; CAD, coronary artery disease; HOMA-IR, homeostatic model assessment for insulin resistance; NT-proBNP, N-terminal pro-B-type natriuretic peptide; and OR, odds ratio.

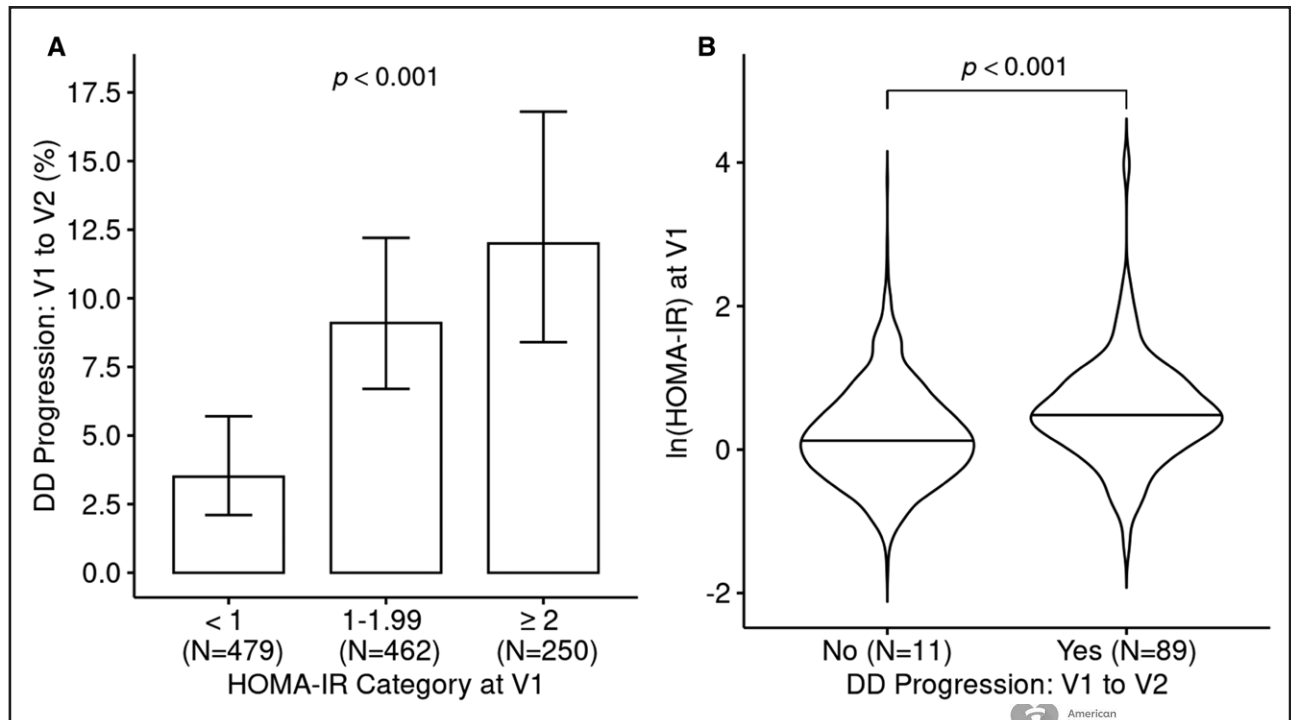


Figure 4. Association between baseline Homeostatic Model Assessment of Insulin Resistance (HOMA-IR) and progression of diastolic dysfunction (DD).

A, Bar graph showing the proportion of participants with progression of DD at visit 2 across categories of baseline HOMA-IR. The percentage of participants with DD progression increased with higher baseline HOMA-IR indices: 3.5% for HOMA-IR <1, 9.1% for HOMA-IR 1–1.99, and 12.0% for HOMA-IR ≥2 ($P < 0.001$; Cochran-Armitage test). **B**, Violin plot showing the distribution of baseline log-transformed HOMA-IR levels by progression status. Participants with DD progression had significantly higher baseline HOMA-IR values ($P < 0.001$; Wilcoxon test).

without clinical outcomes), while cross-sectional imaging studies (MESA,⁷ CARDIA,²³ CHS²⁴) and outcome-only heart failure cohorts (ARIC,⁹ NHANES²⁵) each addressed 1 side of the disease continuum but not both. Uniquely, we show that baseline fasting insulin and HOMA-IR predict progression across the full ordinal 2016 American Society of Echocardiography/European Association of Cardiovascular Imaging diastolic grades on paired echocardiography over only 4 years, and predict long-term mortality and heart failure in the same community population, independent of NT-proBNP and creatinine.

Metabolic Drivers of DD Progression

Although diabetes is a well-established risk factor for HFpEF,²⁶ the contribution of antecedent insulin

resistance to the longitudinal progression of myocardial stiffness has been less clearly defined in community populations. Our study fills this gap by demonstrating that individuals with the highest burden of subclinical insulin resistance are significantly more prone to DD progression. Specifically, rates of DD progression rose from 3.5% in the lowest insulin tertile to 11.6% in the highest. In a multivariable analysis, every 1-SD increase in log-transformed HOMA-IR was associated with a 66% increased odds of DD progression (OR, 1.66 [95% CI, 1.34–2.08]), adjusted for age, creatinine, and NT-proBNP. This implies that the diabetic cardiomyopathy phenotype may indeed commence during the prediabetic phase of insulin resistance.

Mechanistically, our findings align with experimental data suggesting that hyperinsulinemia exerts deleterious

Table 3. Association of Baseline Insulin and HOMA-IR With All-Cause and Cardiovascular Mortality

Biomarker	Outcome	Comparison	HR (95% CI)	P value
Insulin	All-cause mortality	Per 1-SD ↑ log (insulin)	1.11 (1.02–1.21)	0.012
	Cardiovascular mortality	Per 1-SD ↑ log (insulin)	1.16 (1.02–1.33)	0.026
HOMA-IR	All-cause mortality	Per 1-SD ↑ log (HOMA-IR)	1.11 (1.03–1.20)	0.009
	Cardiovascular mortality	Per 1-SD ↑ log (HOMA-IR)	1.16 (1.02–1.32)	0.022

Models adjusted for age, sex, and body mass index. Insulin and HOMA-IR were log-transformed and standardized; hazard ratios reflect risk per 1-SD increase. SD of log (insulin) was 0.65, and SD of log (HOMA-IR) was 0.73. HOMA-IR indicates homeostatic model assessment of insulin resistance; and HR, hazard ratio.

Table 4. Association of Baseline Insulin and HOMA-IR With Heart Failure

Biomarker	BMI strata	Comparison	HR (95% CI)	P value
Insulin	BMI <26	Low Insulin	1.13 (0.67–1.88)	0.650
		High Insulin	2.28 (1.09–4.81)	0.029
	BMI 26–29.9	Low Insulin	1.40 (0.63–3.08)	0.406
		High Insulin	1.69 (0.87–3.27)	0.119
	BMI 30+	Low Insulin	3.93 (1.41–10.92)	0.009
		High Insulin	1.53 (1.01–2.31)	0.043
HOMA-IR	BMI <26	Low HOMA-IR	1.26 (0.75–2.14)	0.380
		High HOMA-IR	2.37 (1.13–4.99)	0.023
	BMI 26–29.9	Low HOMA-IR	1.64 (0.74–3.63)	0.223
		High HOMA-IR	2.15 (1.14–4.06)	0.018
	BMI 30+	Low HOMA-IR	2.95 (0.71–12.21)	0.136
		High HOMA-IR	1.41 (0.93–2.13)	0.108

Models adjusted for age and sex. Low insulin and low HOMA-IR reflect values >1 SD below the mean value on the log scale, while high insulin and high HOMA-IR reflect values more than 1 SD above the mean on the log scale. BMI is reported in kg/m². Reference insulin level reflects values of 3.0–10.6 μ U/mL. Low insulin reflects <3.0 μ U/mL, high insulin reflects >10.6 μ U/mL. The reference level of HOMA-IR reflects values of 0.64–2.77. Low-HOMA-IR reflects <0.64. High HOMA-IR reflects >2.77. BMI indicates body mass index; HOMA-IR, homeostatic model assessment of insulin resistance; and HR, hazard ratio.

effects on the myocardium.^{7,8,27} Insulin is a potent growth factor²⁸ that, when in excess, can stimulate cardiomyocyte hypertrophy²⁸ and a profibrotic state^{8,29} via the mitogen-activated protein kinase pathway.^{30,31} Furthermore, insulin resistance is associated with impaired myocardial energetic efficiency, intramyocardial lipid accumulation (steatosis),³² and inflammation,³³ all of which contribute to increased left ventricular stiffness.

Cardiomyocyte and extracellular matrix stiffening accrue on different timelines. The early, normoglycemic phase captured here likely reflects cardiomyocyte-intrinsic mechanisms (titin hypophosphorylation via blunted NO-cGMP-PKG signaling, impaired calcium handling, lipotoxicity), with interstitial fibrosis a later, less reversible contributor.^{8,29,33} This sequence implies the early window may be more amenable to metabolic intervention than established fibrosis.

The attenuation of this signal when adjusting for BMI (OR, 1.22 [95% CI, 0.95–1.55]) highlights the complex interplay between adiposity and insulin resistance; however, the persistence of the association in other rigorous models suggests that the clock for HFpEF risk starts ticking with the onset of insulin resistance, not just the diagnosis of diabetes.

Notably, the stepwise decline in NT-proBNP across ascending insulin tertiles (Table 1) is concordant with the recognized natriuretic handicap of obesity and insulin resistance,³⁴ in which insulin-induced upregulation of NPR-C-mediated clearance and reduced peptide synthesis lower circulating levels as adiposity rises.³⁵ Clinically, this means natriuretic peptides under-detect the highest metabolic-risk group; consistent with this, adjustment for NT-proBNP strengthened rather than attenuated the insulin-DD association (OR, 1.75; Table 2).

Prognostic Implications of Subclinical Metabolic Disease

Beyond echocardiographic worsening, our secondary analysis of 2030 participants highlights the long-term prognostic importance of these markers. Elevated insulin and HOMA-IR were significantly associated with increased hazards for all-cause (HR, 1.11 per 1-SD) and cardiac mortality (HR, 1.16 per 1-SD). This consistent, monotonic relationship with death affirms that insulin resistance is fundamentally a pathological state. Interestingly, the association with incident heart failure was nonlinear and modified by adiposity. In the normal-weight and overweight strata, the expected direction held, with high HOMA-IR significantly associated with greater heart failure risk in both strata, and high insulin reaching significance in the normal-weight stratum. Among obese participants, this relationship was inverted, and low rather than high insulin conferred the greatest risk in the cohort. This may reflect β -cell burnout,³⁶ in which a low fasting insulin in the setting of established adiposity may signal failed compensation and a more advanced stage of metabolic disease rather than metabolic health, and therefore portends a worse prognosis.^{37,38}

Clinical Implications and Therapeutic Opportunities

Our findings underscore insulin resistance and hyperinsulinemia as modifiable drivers of DD, well before overt hyperglycemia emerges. The stepwise increase in DD progression across HOMA-IR tertiles and the 66% increased odds per 1-SD reinforce the importance of

early risk identification in the prediabetic phase, where intervention may still alter the trajectory toward HFpEF.

Given the independent associations with DD progression and mortality, HOMA-IR and baseline fasting insulin levels could be useful markers for identifying at-risk individuals in the general population. Incorporating insulin resistance into screening strategies allows for earlier intervention, acknowledging that the HFpEF clock starts ticking before diabetes is diagnosed.³⁹

Insulin and HOMA-IR identify at-risk individuals while HbA1c is still normal.⁴⁰ Because HbA1c reflects glycaemia and rises only after beta-cell compensation fails,⁴¹ it cannot capture the earlier hyperinsulinemic euglycemic phase our data implicate in DD progression, and may therefore be an inadequate surrogate.

Pharmacologically, SGLT2i, even in nondiabetics, improve cardiac energetics,⁴² reduce myocardial stiffness and fibrosis, and attenuate adverse remodeling.⁴³ Further, Empagliflozin-induced glycosuria reduces both serum glucose and insulin and promotes ketogenesis,⁴⁴ which also argues for the use of this class of agents in prediabetic patients with elevated baseline HOMA-IR/fasting insulin levels.

GLP-1RA (glucagon-like peptide-1 receptor agonists) and dual GIP (glucose-dependent insulinotropic polypeptide)/GLP-1 agonists reduce epicardial fat⁴⁵ and fibrosis,⁴⁶ improving diastolic parameters regardless of diabetic status.⁴⁶ Metformin may also have a role in this population by activating AMP-activated protein kinase⁴⁷ and improving mitochondrial efficiency,⁴⁷ mechanisms aligned with the prevention of insulin-mediated cardiac remodeling.^{8,31,48}

Weight loss and physical activity remain first-line. Targeted lifestyle interventions improve insulin sensitivity¹² and may directly reverse early DD.⁴⁹

Collectively, our data support a shift toward metabolism-focused prevention, targeting insulin resistance to interrupt the progression from subclinical dysfunction to symptomatic heart failure. Future prospective studies are warranted to determine if early initiation of cardiometabolic pharmacotherapy, combined with lifestyle modification, can arrest this trajectory and improve long-term survival in this at-risk population.

Study Limitations

Our study has several limitations. First, the OCHFSC cohort is predominantly White (98.2%), which may limit generalizability to other racial and ethnic groups with different phenotypes of cardiometabolic risk. Second, while we used paired echocardiography, the assessment of diastolic function is inherently complex; however, we strictly adhered to the 2016 American Society of Echocardiography/European Association of Cardiovascular Imaging guidelines to standardize our definitions.¹⁹ Third, HOMA-IR is a surrogate estimate of insulin

resistance rather than the gold-standard euglycemic clamp, although it remains the standard for large epidemiological studies.⁵⁰ Finally, observational data cannot confirm causality between hyperinsulinemia and myocardial stiffening.

Conclusions

Elevated endogenous insulin and insulin resistance are predictors of DD progression and long-term mortality in the community. These findings suggest that the pathobiology of HFpEF begins with subclinical metabolic dysregulation, identifying a potential therapeutic window for insulin-sensitizing strategies to halt the progression of myocardial stiffness.

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Disclosures

Mayo Clinic has patented chimeric peptides. H.H. Chen reports a modest relationship with UpToDate and holds a patent for designer natriuretic peptides. H.H. Chen reports a significant ownership interest in E-Star Inc. The other authors report no conflicts.

Supplemental Material

Table S1
Figure S1

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